

Mobility Lens Application Report Draft

Name: [replace with your name] BU ID: [replace with your BU ID]
Package: edu.bu.mobilitylens Emulator: [replace with model and API level]

Configuration

Mobility Lens is a Kotlin and Jetpack Compose application that helps a developer think through six ways a phone differs from a stationary computer. The project uses minSdk 24, targetSdk 36, and compileSdk 36. MinSdk 24 means installation is limited to Android 7.0 and newer devices. This is a practical compatibility floor for a small modern Compose application, while still covering a broad set of devices. TargetSdk 36 declares that the app has been tested against Android 16 behavior and lets the platform apply current compatibility and security expectations. CompileSdk 36 supplies the API definitions used while compiling; it does not by itself decide which devices can install the app.

Project Structure

MainActivity is in app/src/main/java/edu/bu/mobilitylens and is the entry point that calls setContent to show the Compose interface. AndroidManifest.xml identifies the application, its launcher activity, and the app label; it requests no unnecessary permissions. The app-level Gradle file applies the Android application plugin, declares the package and SDK settings, enables Compose, and lists the UI dependencies. The version catalog, gradle/libs.versions.toml, centralizes dependency and plugin versions so build settings are easier to read and update. Finally, strings.xml stores user-facing labels and formatted messages, separating reusable text from Kotlin code.

Rotation Observation

[Replace this paragraph after your own emulator test.] I moved to dimension [number/name], typed [example text], and rotated the emulator. The selected dimension and entered text returned to their initial values because this version uses remember state. A rotation normally recreates the activity and its composition, so ordinary remembered state is not restored. An application may need to preserve a draft, the user's current step, or a search query so an interruption such as rotation, a phone call, or memory pressure does not make the person repeat work.

Mobile Design Decision

The screen shows only one mobility dimension at a time and uses clearly labeled Previous and Next buttons. This suits a phone because it reduces visual crowding and supports a short, focused interaction on a small display. A desktop could show all six concepts at once, but on a phone that layout would shrink text and make scanning less comfortable. The app also provides immediate, in-screen validation instead of relying on a separate dialog, keeping attention on the task.

Before submission: replace every bracketed field, run the rotation test yourself, revise this draft in your own voice, and include the required AI and collaboration disclosure as a separate attachment or final section.